Module 1: Logic

Logic is the basis of all mathematical reasoning, and of all automated reasoning. The rules of logic specify the meaning of mathematical statements.

The rules of logic give precise meaning to mathematical statements. These rules are used to distinguish between valid and invalid mathematical arguments. Apart from its importance in understanding mathematical reasoning, logic has numerous applications in Computer Science, varying from design of digital circuits, to the construction of computer programs and verification of correctness of programs.

Proposition:

It is defined as a declarative sentence that is either True or False, but not both.

The **Truth Value** of a proposition is True(denoted as T) if it is a true statement, and False(denoted as F) if it is a false statement.

Example:

1. 2+3=5 (True)
2. ‘z’ is a vowle (False)
3. The sun rises in the East and sets in the West (True)
4. Do you like Mathematics? (?)
5. Go for Lunch (?)
6. X-2=5 (truth value depends on X)

(1), (2), (3) are Propositions or statements whereas (4), (5), (6) are not.

propositional variables are denoted by small letters like p,q,r…

Logical Connectives or Logical operators:

Propositional variables can be combined Logical Connectives or Logical operators to obtain compound statements.

Truth Table

To find the truth value of a proposition, we consider all the possible combinations of the propositions which are joined together by Logical Connectives to form the given compound proposition. This compilation in a tabular format is called a **truth table**.

1. Negation:

If p is a statement then negation of p and is denoted by ~p. and it means the statement “It is not the case that p” or simply “not p”.

The truth table ~P relative to P

p	~p
T	F
F	T

Ex: Give the negation of the following statements:

a) p:

Conjunction (AND)

We can join two statements by “**AND**” operand. It is also known as a conjunction. Its symbolic form is “∧“. In this operator, if anyone of the statement is false, then the result will be false. If both the statements are true, then the result will be true. It has two or more inputs but only one output.

Truth Table for Conjunction (AND)

Logic

Negation

P: A is Even number

~P: A is not an even number

Give the negation of the following statements:

i) P: It is hot.
   ~P: It is not the case that it is hot.

ii) q : 2<3
   ~q: $2 \geq 3$

Conjuction

And

Truth table $p \wedge q$

P	q	$p \bigwedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Ex: Form conjunction of p and q for each of the following

i) p: It is snowing q: I am cold
   $p \wedge q$: It is snowing and I am cold

ii) p: 2<3 q: 3>5

$p \wedge q$: 2<3 and 3>5

2. Conjunction – For any two propositions and , their conjunction is denoted by , which means " and ". The conjunction is True when both and are True, otherwise False.

The truth table of is-

Disjunction

Truth table $pV\ q$ OR

P	q	pVq
T	T	T
T	F	T
F	T	T
F	F	F

Conditional Statements

If P then q

$p \rightarrow q\ or\ p \Rightarrow q$

Truth Table

P	Q	$p \Rightarrow q$

Biconditional

If p and q are statements, then the compound statement " p if and only if q" denoted by $p \Leftrightarrow q$ is called equivalence or biconditional statement.

If both p and q are true or both are false then it is true

Truth Table

P	Q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Tautology

A statement that is true for all possible values of its propositional variables is called Tautology.

A statement that is always false is called contradiction.

The statement which is either true or false is called depending on the truth values of its propositional variable is called a contingency.

Logically Equivalent

If the truth values of two statements are exactly same then those statements are called Logically Equivalent.

Ex: Make a truth table for

1. $(\sim p \wedge q) \vee p$

P	Q	$\sim p$	$\sim p \wedge q$	$(\sim p \wedge q) \vee p$		
T	T	F	F	T		
T	F	F	F	T		
F	T	T	T	T		
F	F	T	F	F		

2. $(p \vee q) \vee \sim q$

P	Q	$p \vee q$	$\sim q$	$(p \vee q) \vee \sim q$		
T	T	T	F	T		
T	F	T	T	T		
F	T	T	F	T		

F	F	F	T	T		

This Statement is Tautology

1. $(p \Rightarrow q) \Leftrightarrow (\sim q \Longrightarrow \sim p)$

P	Q	$p \Rightarrow q$	~q	~P	~q$\Longrightarrow$ $\sim p$		
T	T	T	F	F	T	T	
T	F	F	T	F	F	T	
F	T	T	F	T	T	T	
F	F	T	T	T	T	T	

This Statement is Tautology

2. $p \wedge \sim p$

P	~p	1. $p \wedge \sim p$
T	F	F
T	F	F
F	T	F
F	T	F

1. $(p \rightarrow q) \wedge (p \vee q)$

P	q	$p \rightarrow q$	$p \vee q$	$(p \rightarrow q) \wedge (p \vee q)$
T	T	T	T	T
T	F	F	T	F
F	T	T	T	T
F	F	T	F	F

EQUIVALENCE

If P and Q ARE TWO STATEMENTS THEN WE SAY THAT P IS EQUIVALENT TO Q if the truth tables of statement P and Q are exactly same and is denoted by $P \equiv Q$

Example: $P \vee P \equiv P$ $\quad P \wedge P \equiv P$

$p \Longrightarrow q \equiv (\sim p \vee q)$

P	Q	$p \rightarrow q$	~p	$\sim p \vee q$	
T	T	T	F	T	
T	F	F	F	F	
F	T	T	T	T	
F	F	T	T	T	
		(1)		(2)	

From column (1) and (2) in the truth table $p \Longrightarrow q \equiv (\sim p \vee q)$

Example: 2 $\quad p \rightarrow q \equiv (\sim q \rightarrow \sim p)$

P	Q	$p \rightarrow q$	~p	$\sim q$	$\sim q \rightarrow \sim$p
T	T	T	F	F	T
T	F	F	F	T	F
F	T	T	T	F	T
F	F	T	T	T	T
		(1)			(2)

Example:1. $(p \leftrightarrow q) \equiv ((p \rightarrow q) \wedge (q \rightarrow p))$

2. $\sim(p \leftrightarrow q) \equiv ((p \wedge \sim q) \vee (q \wedge \sim p))$
3. $(p \vee q) \vee r \equiv p \vee (q \vee r)$

P	q	r	~p	$\sim q$	$\sim q \rightarrow \sim$p
T	T	T			
T	T	F			
T	F	T			
T	F	F			
F	T	T			
F	T	F			
F	F	T			
F	F	F			

Laws of Algebra of Proposition:

1. Idempotent law: $P \vee P \equiv P \qquad P \wedge P \equiv P$
2. Identity Law: $P \vee F \equiv P \qquad P \wedge T \equiv P$
3. Complement law: $P \vee \sim P \equiv T \qquad P \wedge \sim P \equiv F$
4. Commutative law: $p \vee q \equiv q \vee p \qquad p \wedge q \equiv q \wedge p$
5. Associative law: $(p \vee q) \vee r \equiv p \vee (q \vee r)$
   $(p \wedge q) \wedge r \equiv p \wedge (q \wedge r)$
6. Distributive Law: $p \vee (q \wedge r) \equiv (p \vee q) \wedge (p \vee r)$
   $p \wedge (q \vee r) \equiv (p \vee q) \wedge (p \vee r)$
7. Absorption Law: $p \vee (p \wedge q) \equiv p \qquad p \wedge (p \vee q) \equiv p$
8. DeMorgan's Law: $\sim(p \vee q) \equiv \sim p \wedge \sim q$

$$\sim(p \wedge q) \equiv \sim p \bigvee \sim q$$

Normal form:

There are two types of Normal forms

1. Disjunctive Normal form (DNF) (Sum)
2. Conjunctive Normal form (CNF) (Product)

The statements

p,~p, $p \wedge \sim p$, $\sim p \wedge q, p \wedge \sim q, \sim p \wedge \sim q$ are called elementary product in two variable

q, ~q, p$\vee$ $q, p \vee \sim q, \sim p \vee \sim q$ are called elementary sum in 2 variables.

A compound statement which consists of a **sum of elementary products** and which is equivalent to a given proposition is called Disjunctive normal form. (DNF)

Ex: $(\sim p \wedge q) \vee (\sim p \wedge \sim q)$

A compound statement which consists of a **product of elementary sums** and which is equivalent to a given proposition is called Conjunctive normal form. (CNF)

Ex: (p$\vee$ $q) \wedge (\sim p \vee q$)

Procedure to obtain DNF and CNF

Step 1: If the connectives $\rightarrow and \leftrightarrow$ are present in the given formula then replace those by $\wedge$,$\vee$, $\sim$

$P \rightarrow q$ is replaced by $\sim p \vee q$ and

$p \leftrightarrow q$ is replaced by either $(p \wedge q) \vee (\sim p \wedge \sim q)$

Or $(\sim p \vee q) \wedge (\sim q \vee p)$

Step 2: If negation is present before the given proposition then DeMorgan's Laws are applied to brought it before the variable

Step 3: If required, then distributive laws and idempotent laws are applied.

Step 4: If DNF is equivalent to F then it is omitted and if CNF is equivalent to T is omitted.

Example 1: Find the disjunctive normal form of the following statements:

1. $\neg(\neg(p \leftrightarrow q) \wedge r)$

Solution: $\neg(\neg(p \leftrightarrow q) \wedge r) \equiv \neg(\neg((p \wedge q) \vee (\neg p \wedge \neg q) \wedge r)$

$$\equiv \neg((\neg(p \wedge q) \wedge \neg(\neg p \wedge \neg q)) \wedge r)$$

….(By using DMorgan's Law

$$\equiv \neg((\neg p \vee \neg q) \wedge (p \vee q) \bigwedge r)$$

….(By using DMorgan's Law)

$$\equiv \neg[((\neg p \wedge p) \vee (\neg p \wedge q) \vee (\neg q \wedge p) \vee (\neg q \wedge q)) \wedge r]$$

…..(By Distributive Law)

$$\equiv \neg[((\neg p \wedge q) \vee F \vee (\neg q \wedge p) \vee F) \wedge r]..$$

$$\equiv \neg[((\neg p \wedge q) \vee (\neg q \wedge p)) \wedge r]$$

$$\equiv \neg\big((\neg p \wedge q) \vee (\neg q \wedge p)\big) \vee \neg r$$

….(By DeMorgan;s Law)

$$\equiv \neg((\neg p \vee \neg q) \wedge (\neg p \vee p) \wedge (q \vee \neg q) \wedge (q \vee p)) \vee \neg r$$

$$\equiv \neg((\neg p \vee \neg q) \wedge (q \vee p)) \vee \neg r$$

$$\equiv (\neg(\neg p \vee \neg q) \vee \neg(q \vee p)) \vee \neg r$$

$$\equiv (p \wedge q) \vee (\neg q \wedge \neg p)) \vee \neg r$$

$Ex2: p \vee (\neg p \rightarrow (q \vee (q \rightarrow \neg r)))$

$Solution: p \vee \Big(\neg p \rightarrow \big(q \vee (q \rightarrow \neg r)\big)\Big) \equiv p \vee \Big(\neg p \rightarrow \big(q \vee (\neg q \vee \neg r)\big)\Big)$

$$\equiv p \vee (p \vee (q \vee (\neg q \vee \neg r)))$$

$$\equiv p \vee p \vee q \vee \neg q \vee \neg r$$

$$\equiv p \vee q \vee \neg q \vee \neg r$$

$$\equiv p \vee T \vee \neg r$$

$$\equiv p \vee \neg r$$

Find the conjunctive normal forms of the following statements:

i) $(p \wedge \sim(q \wedge r) \vee (p \rightarrow q))$

Solution:

$(p \wedge \sim(q \wedge r) \vee (p \rightarrow q)) \equiv (p \wedge (\sim q \vee \sim r) \vee (\sim p \vee q))$ ….(DeMorgan's Law)

$\equiv (p \wedge \sim q) \vee (p \wedge \sim r) \vee (\sim p \vee q))$ ..(Distributive Law)

$\equiv (p \vee p) \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim r) \vee (\sim p \vee q)$ .. ..(Distributive Law)

$\equiv p \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (\sim q \vee \sim r \vee \sim p \vee q)$

$\equiv p \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (\sim q \vee q \vee \sim r \vee \sim p)$

$$\equiv p \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (T \vee \sim r \vee \sim p)$$
$$\equiv p \wedge (p \vee \sim r) \wedge (\sim q \vee p) \wedge (\sim r \vee \sim p)$$

ii) $(p \wedge \sim (q \vee r)) \vee (((p \wedge q) \vee \sim r) \wedge p)$

$Solution$: $(p \wedge \sim (q \vee r)) \vee (((p \wedge q) \vee \sim r) \wedge p)$

$\equiv (p \wedge (\sim q \wedge \sim r)) \vee (((p \vee \sim r) \wedge (q \vee \sim r)) \wedge p)$
…(Demorgan's Law & Distributive law
$\equiv (p \wedge (\sim q \wedge \sim r)) \vee (((p \vee \sim r) \wedge (q \vee \sim r)) \wedge p)$
$\equiv (p \wedge (\sim q \wedge \sim r)) \vee ((p \wedge (p \vee \sim r) \wedge (q \vee \sim r)))$
$\equiv (p \wedge (\sim q \wedge \sim r)) \vee (p \wedge (q \vee \sim r))$ …absorption law
$\equiv ((p \wedge (\sim q \wedge \sim r)) \vee p) \wedge ((p \wedge \sim q \wedge \sim r) \vee q \vee \sim r)$ …absorption law
$\equiv p \wedge (((p \wedge \sim q) \wedge \sim r) \vee \sim r) \vee q)$
$\equiv p \wedge (\sim r \vee q)$
Absorption law: $p \wedge (p \vee q) \equiv p \quad p \vee (p \wedge q) \equiv p$

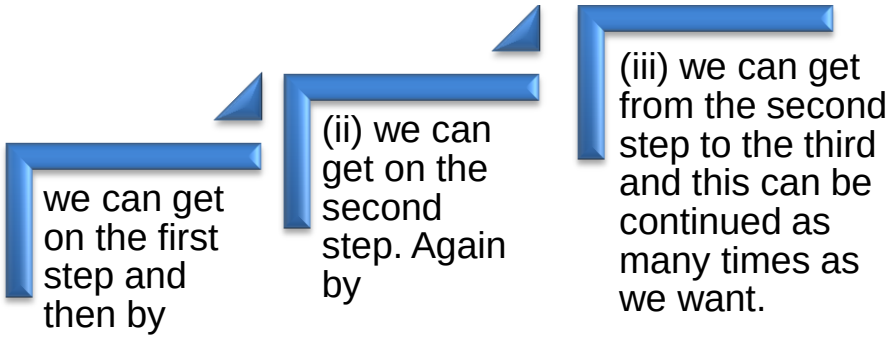
iii) Find Disjunctive normal form of $(\sim p \rightarrow q) \wedge (q \leftrightarrow p)$

Solution: $(\sim p \rightarrow q) \wedge (q \leftrightarrow p) \equiv (p \vee q) \wedge (() \vee ())$

Mathematical Induction

The technique of mathematical induction is frequently used in proving results in mathematics and computer science. The term mathematical induction was first used by De-Morgan.

Suppose we have a ladder of infinite steps. How can we prove that we can climb to any step? We can do so if, we get on the next step. Because of



This is the general approach of Mathematical Induction.

Proof by Mathematical Induction:

The proof consists of three parts.

Let p(n) denote a statement (a formula or a theorem) associated with n = 1, 2, 3, any natural number.

Step I (Verification): We first verify that P(n) i.e. the theorem is true for n=1

Step II (Induction Property): Assuming P(n) i.e. the theorem to be true for some value n = k, say, then we prove by some known method that is true for n = k + 1 i.e., we can get from kth step to (k+1)th step of the ladder.

Step III (Inductive Property) : The result is true for any value of $n \in N$.

Example 2: Prove n < 2n for all positive integers n.

Solution:

Step 1: Putting n = 1, 2, we see that $1 < 2*1 = 2$, $2 < 2*2 = 4$

The result is true for n = 1, 2. P(1), P(2) are true.

Step 2: Assume that the result is true for n = k i.e., assume that P(k) is true. Therefore, $k < 2k$

Now, multiply both sides by 2

Therefore the result is true for n = k+1.

Step 3 : From step I and II, we conclude that the result is true for any positive integer

Example 3 : Use Mathematical Induction to show that

$1+5+9+\ldots\ldots\ldots\ldots+(4n-3) = n(2n-1)$

Solution:

Step 1: Let P(n) be the statement $1+5+9+\ldots\ldots\ldots\ldots+(4n-3) = n(2n-1)$

Will prove for n= 1 and 2,

P(1): $1= 1(2-1)=1$ is true,

P(2): $1+5 = 2(2*2-1)$, $6=6$

Assume it is true for P(k) so P(k) : $1+5\ldots..+(4k-3)=k(2k-1)$----(1)

Step 2: The next term k+1 will be $1+5\ldots..+(4k-3)+(4(k+1)-3)$

Substituting equation 1 it will be $=k(2k-1)+(4(k+1)-3)=K(2k)-k+4k+1=2k2+3k+1 \quad =(k+1)(2k+1)$

RHS= $(k+1)(2(k+1)-1)= (k+1)(2k+1)$

Therefore the result is true for n = k+1.

Step 3: From step 1 and 2, we conclude that the result is true for any positive integer.

Example 4: 3^n-1 is a multiple of 2 for n = 1, 2, ...

Solution:

Step 1: For n=1, 3^1-1 =3−1=2, which is a multiple of 2

Step 2: Let us assume 3^n-1 is true for n=k, Hence, 3^K-1

is true (It is an assumption)

Step 3: We have to prove that $3^{K+1}-1$ is also a multiple of 2

$3^{K+1}-1$ =3×3^K-1 =(2×3^K)+(3^K-1)

The first part (2×3^K) is certain to be a multiple of 2 and the second part (3^K-1) is also true as our previous assumption.

Hence, $3^{K+1}-1$ is a multiple of 2.

So, it is proved that 3^n-1 is a multiple of 2.